

AI in Genomics 2026

Tunisia · 14–18 Sept 2026

Practical 3: Compare

Presenter: Yu Sugihara

PROGRAMME

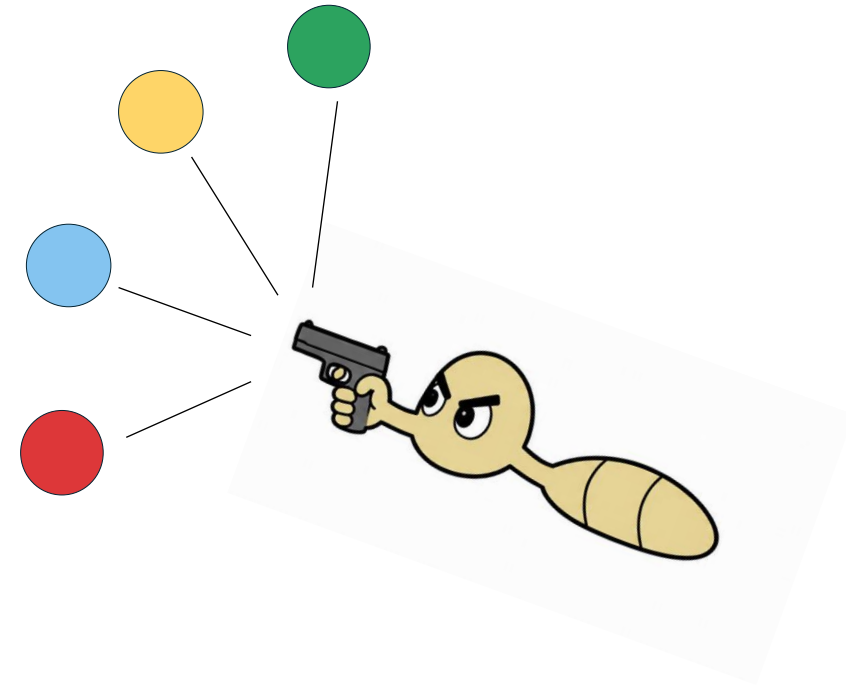


Pyricularia oryzae secretes virulence-associated proteins to infect rice

Rice blast disease



Pyricularia oryzae
spores



Rice immune receptor Pikm recognizes distinct pathogen-secreted proteins



Pikm

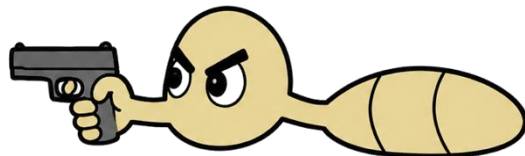
All three strains were inoculated on rice carrying Pikm



Wild type



No AVR protein to recognize



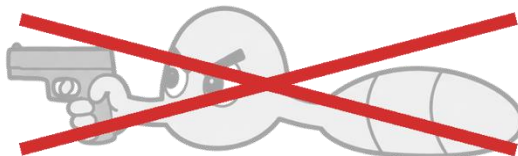
Rice infected



AVR-PikD



Recognized by Pikm



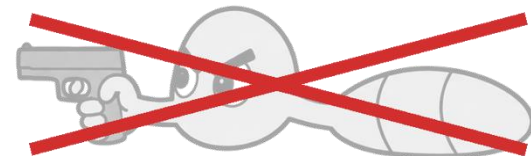
Rice protected



AVR-Mgk1



Recognized by Pikm



Rice protected

AVR-Mgk1 is unrelated to AVR-PikD in sequence

AVR-Mgk1
MWSKSTIVTSAALLFNQAALVAARNCRIWQDMGSV-----WQEVVVVTPPVTVDIITKR-----HGAFSLFVPVGCG---IRDTGGALRATETDDPW

AVR-PikD
M---RVTTFTNTFLTLGTVAVVNAETGNKYIEKRAIDL SRERDPNFFDHGP IVP P ECFWFMF KNNVRQDAGTCYSSWKMDMKVGP NWVHIKSDDNCNL SGDFPPGW IVLGKKRP GF

Amino acid sequence similarity is only 10%

BBQ (Big Biological Question)

How does an immune receptor recognize two distinct proteins?

GoHREP

G Goal

Find out how one rice immune receptor recognizes two pathogen proteins with no sequence similarity

GoHREP

G Goal

Find out how one rice immune receptor recognizes two pathogen proteins with no sequence similarity

H Hypothesis

AVR-PikD and AVR-Mgk1 share the same fold

GoHREP

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H Hypothesis

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R Rationale

The rice immune receptor Pia recognizes effectors that share a fold but not a sequence

GoHREP

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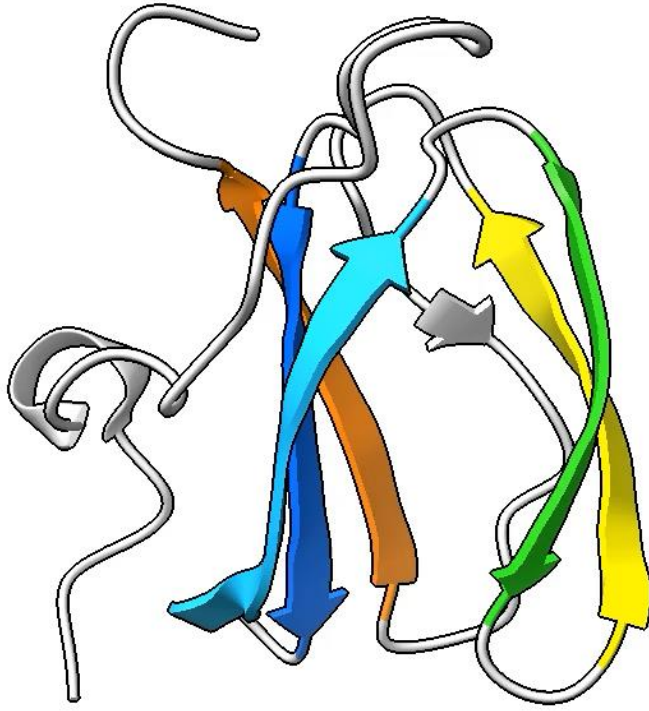
The rice immune receptor Pia recognizes effectors that share a fold but not a sequence

E Experimental plan

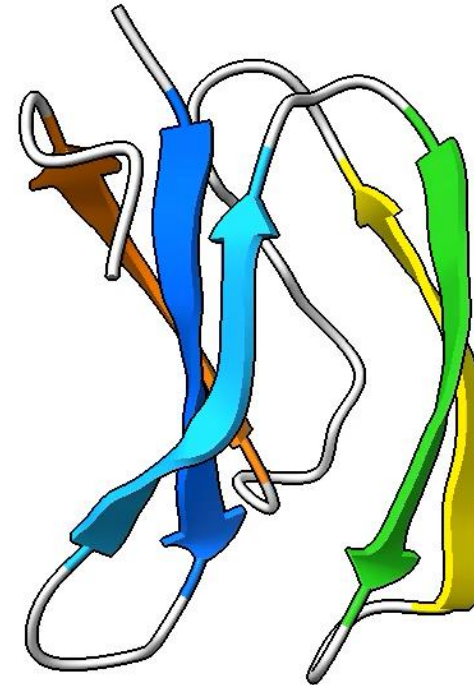
Predict the AVR-Mgk1 structure with AlphaFold 2 and compare it with the published AVR-PikD structure

AVR-PikD and AVR-Mgk1 share the MAX fold despite having no sequence similarity

AVR-PikD

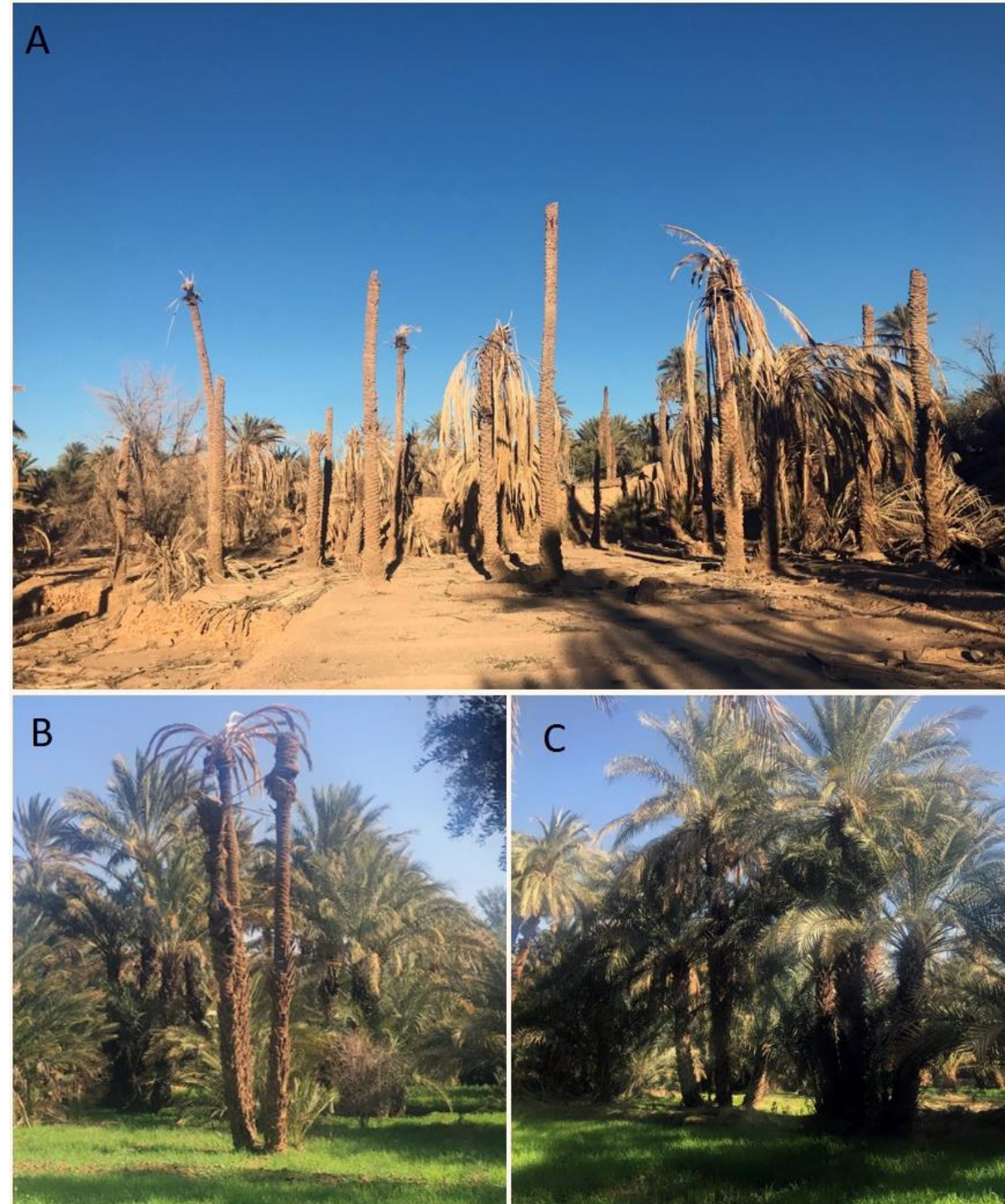


AVR-Mgk1



***Fusarium oxysporum* f. sp. *albedinis*
causes bayoud disease of date palm**

Date palm affected by bayoud disease
in a natural oasis in Morocco
(Rafiqi et al., 2022 · PMID 35617325)



Structural clustering links distinct sequence-based tribes in 305 *Fusarium* dataset

1

305 *Fusarium* genome annotations



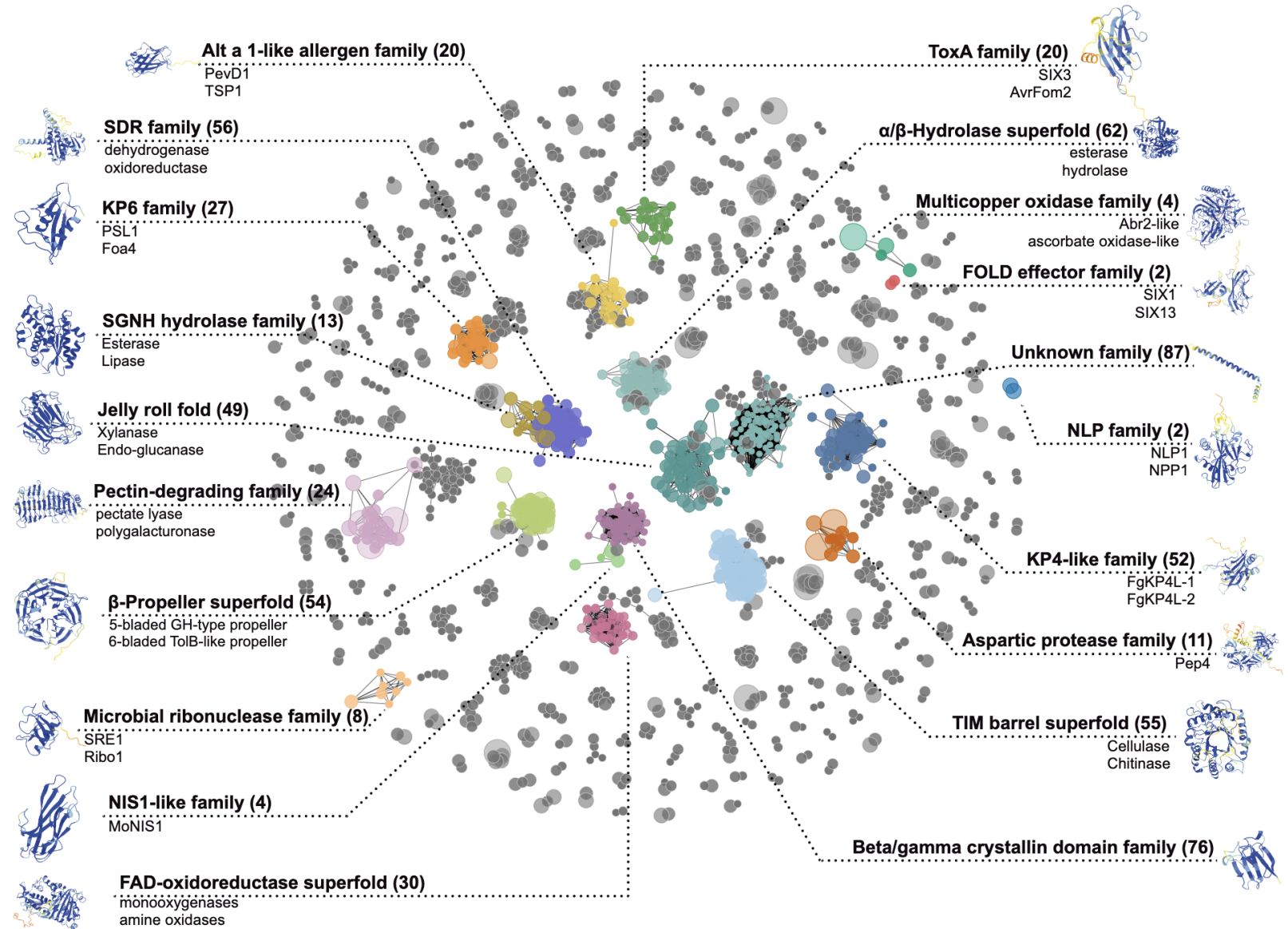
2

Sequence-based clustering



3

Structure-based clustering



Exercise 1: Comparing sequences

- 1.1 Search the NCBI database with BLAST
- 1.2 Search your own sequences with BLAST
- 1.3 Domain annotation with InterProScan

Exercise 2: Comparing protein structures

- 2.1 Visualizing structures with ChimeraX
- 2.2 Aligning structures with TM-align

Exercise 3: Searching for similar structures with Foldseek

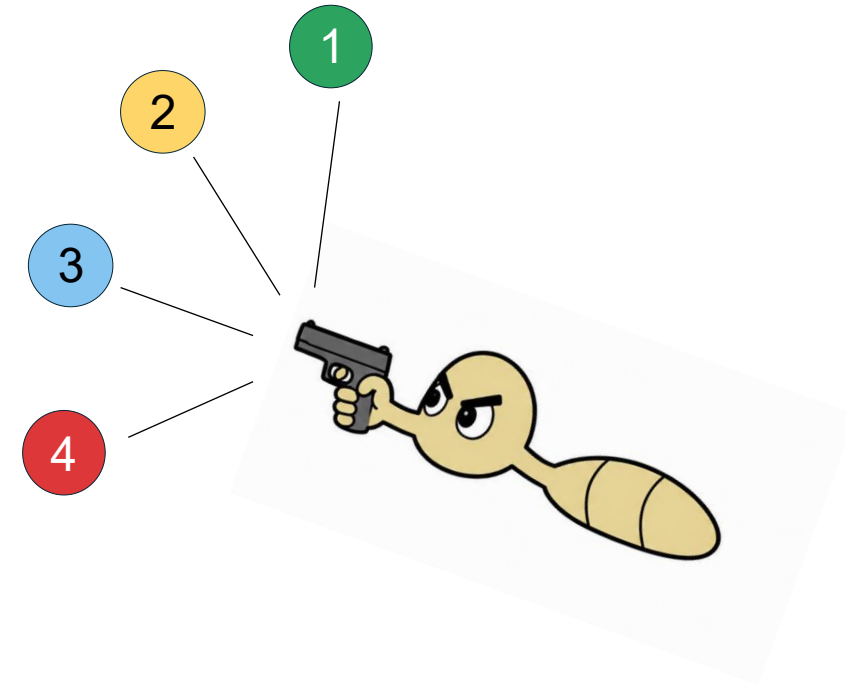
Exercise 4: Aligning many structures with FoldMason

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Analyze the sequences before the structures

1. Did you try more than one database?
2. Did you relax the parameters to find distant relatives?
3. Did you check the sequences for domain annotations?

Exercise 1.1 and 1.2: BLAST (Dataset 1)

1.1 Search the NCBI database

BLAST® » blastp suite [Home](#)

blastn **blastp** blastx tblastn tblastx

Standard Protein BLAST

BLASTP programs search protein databases using a protein query. more...

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#) Query subrange [?](#)

From

To

Or, upload file No file selected. [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#) **leave this unchecked**

Choose Search Set

Database **Non-redundant protein sequences (nr)** [?](#) **choose the database**

Organism Optional ☐ exclude

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown. [?](#)

Exclude Optional ☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm

☐ Quick BLASTP (Accelerated protein-protein BLAST)

☒ blastp (protein-protein BLAST)

☐ PSI-BLAST (Position-Specific Iterated BLAST)

☐ PHI-BLAST (Pattern Hit Initiated BLAST)

☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm [?](#)

4 Search database nr using Blastp (protein-protein BLAST)

☐ Show results in a new window

1.2 Search your own sequences

BLAST® » blastp suite

blastn **blastp** blastx tblastn tblastx **Align Sequences Protein BLAST**

BLASTP programs search protein subjects using a protein query. more...

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#) Query subrange [?](#)

From

To

Or, upload file No file selected. [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☒ Align two or more sequences [?](#) **1**

Enter Subject Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#) Subject subrange [?](#)

From

To

Or, upload file No file selected. [?](#) **3**

Program Selection

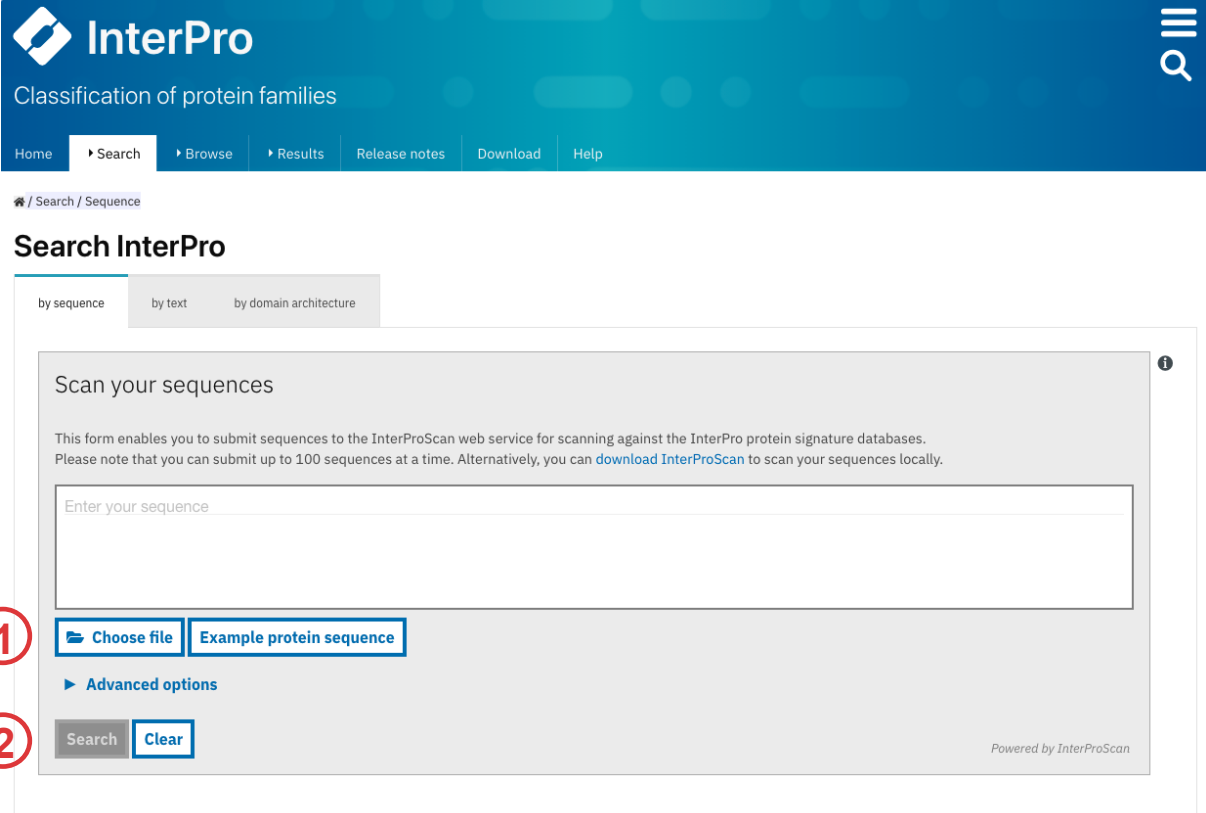
Algorithm ☒ blastp (protein-protein BLAST)

Choose a BLAST algorithm [?](#)

4 Search protein sequence using Blastp (protein-protein BLAST)

☒ Show results in a new window

Exercise 1.3: Domain annotation with InterProScan (Dataset 1)



The screenshot shows the InterPro website's search interface. At the top is a blue header with the InterPro logo and the tagline 'Classification of protein families'. Below the header is a navigation bar with links: Home, Search, Browse, Results, Release notes, Download, and Help. The 'Search' link is highlighted. Below the navigation bar, the breadcrumb 'Search / Sequence' is visible. The main heading is 'Search InterPro'. Underneath, there are three tabs: 'by sequence' (selected), 'by text', and 'by domain architecture'. The 'by sequence' tab contains a form titled 'Scan your sequences'. The form includes a text input field labeled 'Enter your sequence'. Below the input field are two buttons: 'Choose file' and 'Example protein sequence'. Below these buttons is a link 'Advanced options'. At the bottom of the form are two buttons: 'Search' and 'Clear'. A red circle with the number '1' is next to the 'Choose file' button, and a red circle with the number '2' is next to the 'Search' button. The footer of the form says 'Powered by InterProScan'.

InterPro
Classification of protein families

Home Search Browse Results Release notes Download Help

Search / Sequence

Search InterPro

by sequence by text by domain architecture

Scan your sequences

This form enables you to submit sequences to the InterProScan web service for scanning against the InterPro protein signature databases. Please note that you can submit up to 100 sequences at a time. Alternatively, you can [download InterProScan](#) to scan your sequences locally.

Enter your sequence

1 Choose file Example protein sequence

Advanced options

2 Search Clear

Powered by InterProScan

<https://www.ebi.ac.uk/interpro/search/sequence/>

An all-vs-all BLAST search finds similarity only between Protein 1 and Protein 2

[Download](#) [Graphics](#)

Protein_1

Sequence ID: **Query_1396600** Length: **91** Number of Matches: **1**

Range 1: 1 to 91 [Graphics](#)

[Next Match](#) [Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps
190 bits(482)	2e-69	Compositional matrix adjust.	91/91(100%)	91/91(100%)	0/91(0%)
Query 1	DLATGDLHHSAVCVTKRVNFFGHKTFQISPNATMCACTFYKGRNTGTNPWDKCEDCKYDG 60				
Sbjct 1	DLATGDLHHSAVCVTKRVNFFGHKTFQISPNATMCACTFYKGRNTGTNPWDKCEDCKYDG 60				
Query 61	LKCTSENGTIGGDEFKYYCEKKCGAKGSEAD 91				
Sbjct 61	LKCTSENGTIGGDEFKYYCEKKCGAKGSEAD 91				

[Download](#) [Graphics](#)

Protein_2

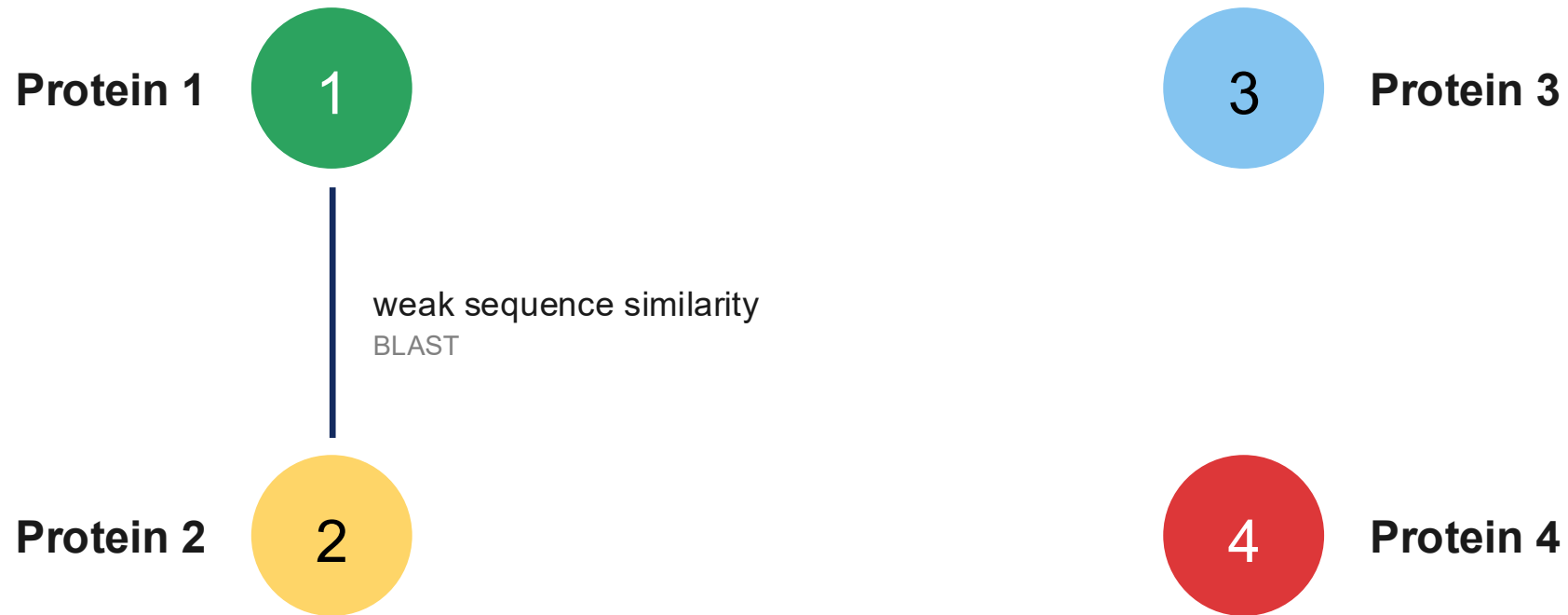
Sequence ID: **Query_1396601** Length: **91** Number of Matches: **1**

Range 1: 2 to 88 [Graphics](#)

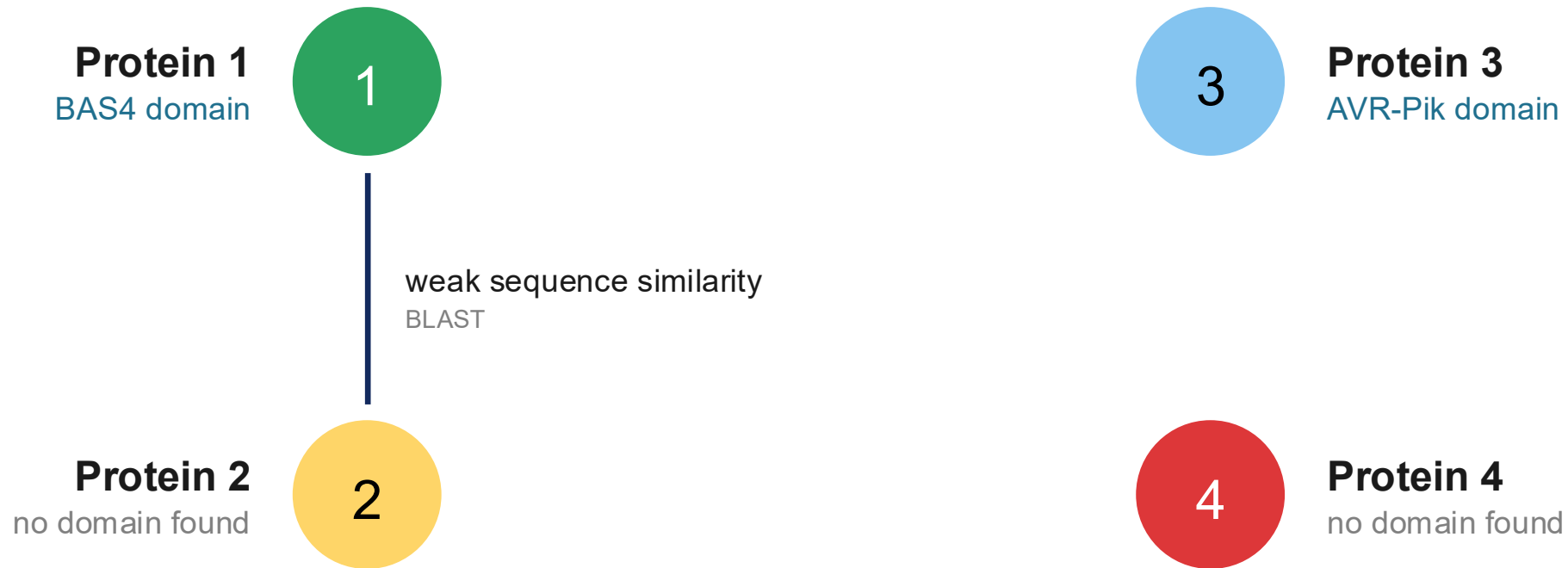
[Next Match](#) [Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps
30.8 bits(68)	9e-07	Compositional matrix adjust.	15/87(17%)	37/87(42%)	0/87(0%)
Query 2	LATGDLHHSAVCVTKRVNFFGHKTFQISPNATMCACTFYKGRNTGTNPWDKCEDCKYDGL 61				
Sbjct 2	IEEGNYCCAGACISSVTVVGVSYSYTKNIAGTKCACELMRALDSGSAKGDAPNAYFNGE 61				
Query 62	KCTSENGTIGGDEFKYYCEKKCGAKGS 88				
Sbjct 62	NCLDPDCNLSYSRLRHFCVNECGCAGA 88				

An all-vs-all BLAST search finds similarity only between Protein 1 and Protein 2



InterProScan annotates Protein 1 and Protein 3 with two different domains



Gene names can be misleading

BLAST search for Protein_1, against nr

Sequences producing significant alignments

Download

Select columns

Show

100

?

☒ select all 100 sequences selected

GenPept

Graphics

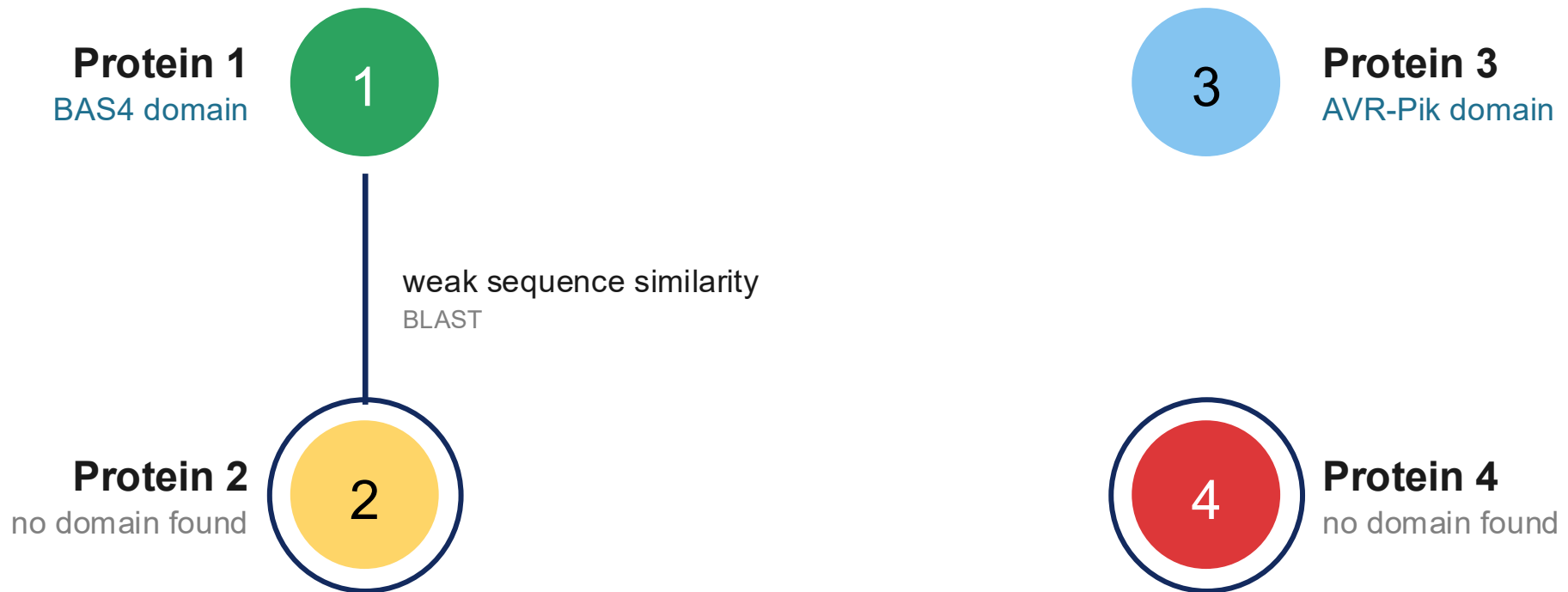
Distance tree of results

Multiple alignment

MSA Viewer

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	uncharacterized protein MGG_02154 [Pyricularia oryzae 70-15]	Pyricularia oryzae 70-15	192	192	100%	5e-61	100.00%	108	XP_003708926.1
☒	hypothetical protein PoMZ_09164 [Pyricularia oryzae]	Pyricularia oryzae	190	190	100%	2e-60	98.90%	108	QBZ53484.1
☒	hypothetical protein MCOR20_004219 [Pyricularia oryzae]	Pyricularia oryzae	186	186	98%	2e-58	97.75%	120	KAI6411550.1
☒	hypothetical protein MCOR26_005222 [Pyricularia oryzae]	Pyricularia oryzae	185	185	100%	8e-58	96.70%	140	KAI6277277.1
☒	uncharacterized protein PpBr36_10495 [Pyricularia pennisetigena]	Pyricularia pennisetigena	168	168	100%	1e-51	83.52%	108	XP_029743940.1
☒	hypothetical protein PspLS_11087 [Pyricularia sp. CBS 133598]	Pyricularia sp. CBS 133598	158	158	100%	2e-47	80.22%	108	TLD14715.1
☒	hypothetical protein MCOR25_003031 [Pyricularia grisea]	Pyricularia grisea	139	139	99%	4e-40	68.89%	108	KAI6375137.1
☒	hypothetical protein PgNI_05328 [Pyricularia grisea]	Pyricularia grisea	137	137	99%	1e-39	67.78%	108	XP_030983304.1
☒	uncharacterized protein CTRU02_207806 [Colletotrichum truncatum]	Colletotrichum truncatum	113	113	98%	7e-30	59.55%	104	XP_036574883.1
☒	hypothetical protein CH63R_07242 [Colletotrichum higginsianum IMI 349063]	Colletotrichum higginsianum IMI 349063	111	111	99%	6e-29	52.94%	116	XP_018156995.1
☒	hypothetical protein QIS74_02669 [Colletotrichum tabaci]	Colletotrichum tabaci	111	111	99%	6e-29	52.94%	116	KAK6224342.1
☒	hypothetical protein FAUST_4719 [Fusarium austroamericanum]	Fusarium austroamericanum	110	110	100%	1e-28	51.96%	115	KAF5239834.1
☒	hypothetical protein FGSG_09127 [Fusarium graminearum]	Fusarium graminearum	110	110	100%	2e-28	50.98%	115	YEN40579.1
☒	hypothetical protein FGSG_09127 [Fusarium graminearum PH-1]	Fusarium graminearum PH-1	109	109	100%	3e-28	50.98%	115	XP_011328661.1
☒	hypothetical protein ACYAQ0_007035 [Fusarium cf. solani]	Fusarium cf. solani	109	109	98%	4e-28	51.00%	116	KAN9084925.1
☒	hypothetical protein FSPOR_5148 [Fusarium sporotrichioides]	Fusarium sporotrichioides	109	109	100%	4e-28	51.96%	115	RGP68678.1
☒	uncharacterized protein NECHADRAFT_77130 [Fusarium vanettenii 77-13-4]	Fusarium vanettenii 77-13-4	109	109	98%	4e-28	51.00%	116	XP_003044014.1

How do we characterize a protein with no known domain?



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Exercise 4: Aligning many structures with FoldMason

Exercise 2.1: Visualizing structures with ChimeraX (Dataset 2)

1. Open the PDB files from Dataset 2 in ChimeraX.
2. Superpose every structure onto #1 with matchmaker.

```
mm #2 to #1  
mm #3 to #1  
mm #4 to #1
```

or

```
mm #2-4 to #1
```

1_BAS4 and 2_IF superpose almost perfectly



Structure vs Architecture vs Fold

Similar structure



Exterior and floor plan both match

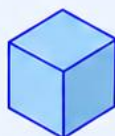


Similar structure = strong 3D match
Good superposition

Similar architecture

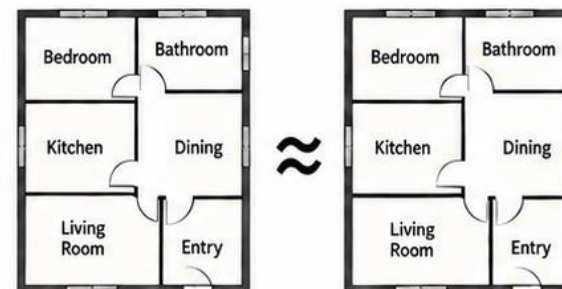


Floor plan may be the same or different

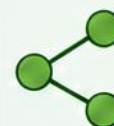


Similar architecture = similar overall 3D layout
Connectivity not specified

Similar fold (topology)



Same connectivity / floor plan

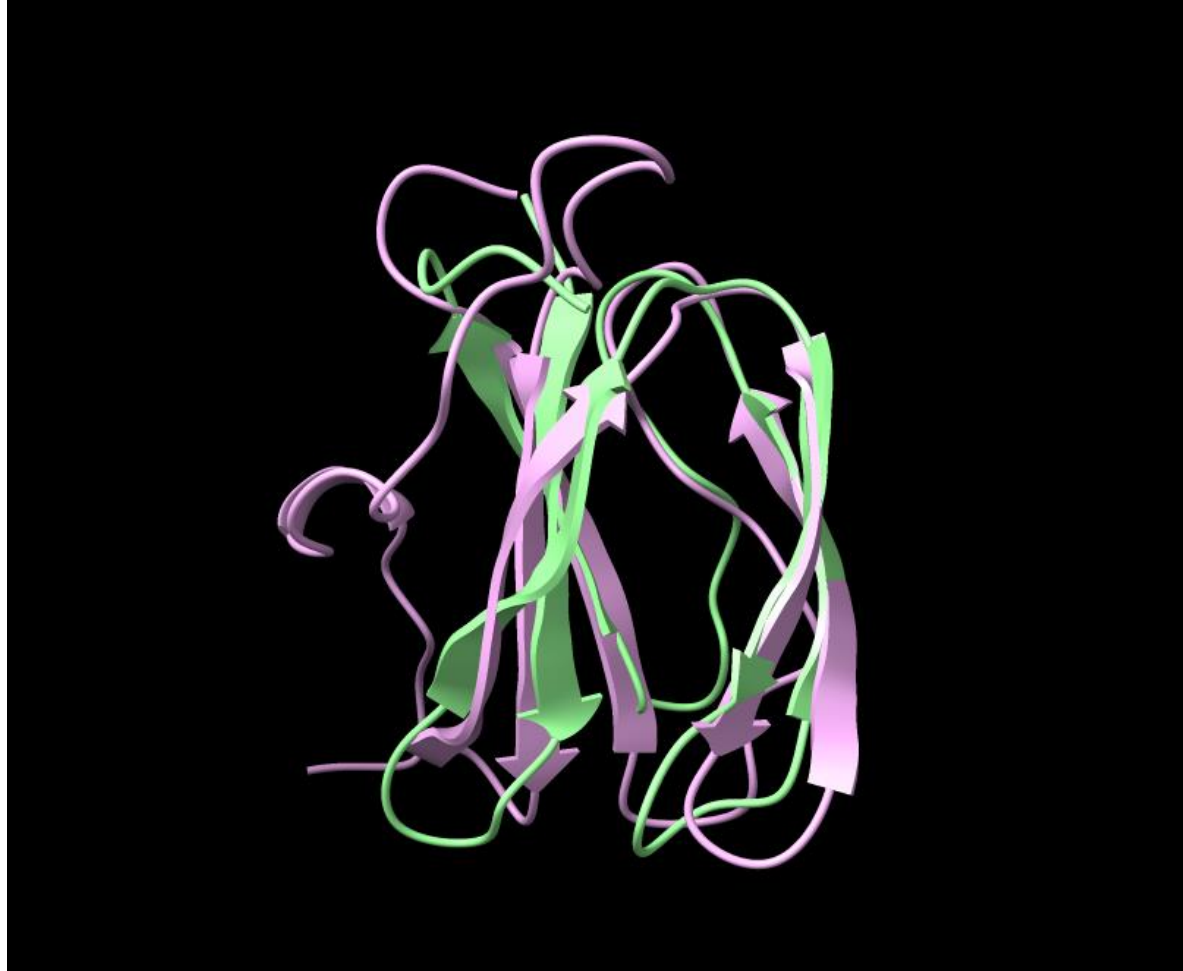


Similar fold = similar arrangement + connectivity
How parts are connected



Key point: structure is the strongest match; architecture and fold capture different aspects of similarity.

Do 3_AVR-PikD and 4_AVR-Mgk1 share a fold?



Exercise 2.2: Aligning structures with TM-align (Dataset 2)

Upload the structures from Dataset 2 to TM-align:

<https://zhanggroup.org/TM-align/>

Upload the first structure ①

Upload the second structure ②

TM-align on-line (view an example of output)

Note: This server is only for pair-wise structure comparison. If you want to match one protein structure with all proteins in the PDB library, you can do it in [COFACTOR Server](#).

- Input Structure 1 in [PDB format](#) or [PDBx/mmCIF format](#) (mandatory):
Please copy and paste your structure file here. [Sample input](#)

Or upload the stucture file:
 No file selected.

- Input Structure 2 in [PDB format](#) or [PDBx/mmCIF format](#) (mandatory):
Please copy and paste your structure file here. [Sample input](#)

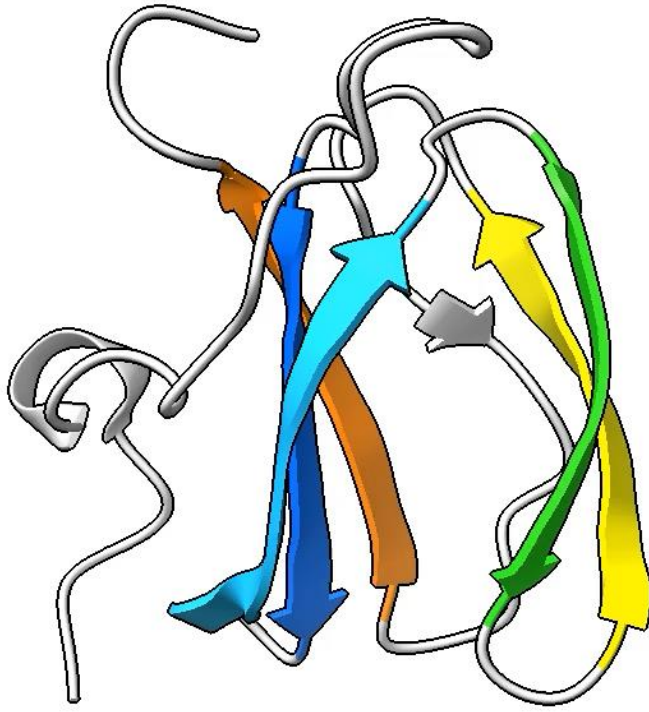
Or upload the stucture file:
 No file selected.

- Input Email: (optional, where results will be sent to)

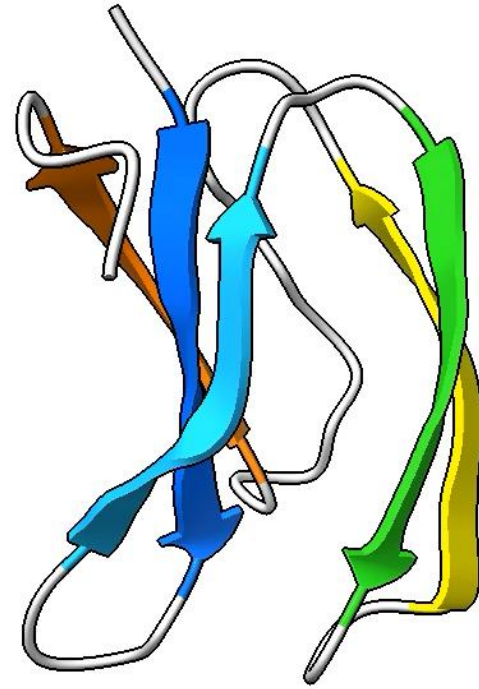
-

AVR-PikD and AVR-Mgk1 share the MAX fold despite having no sequence similarity

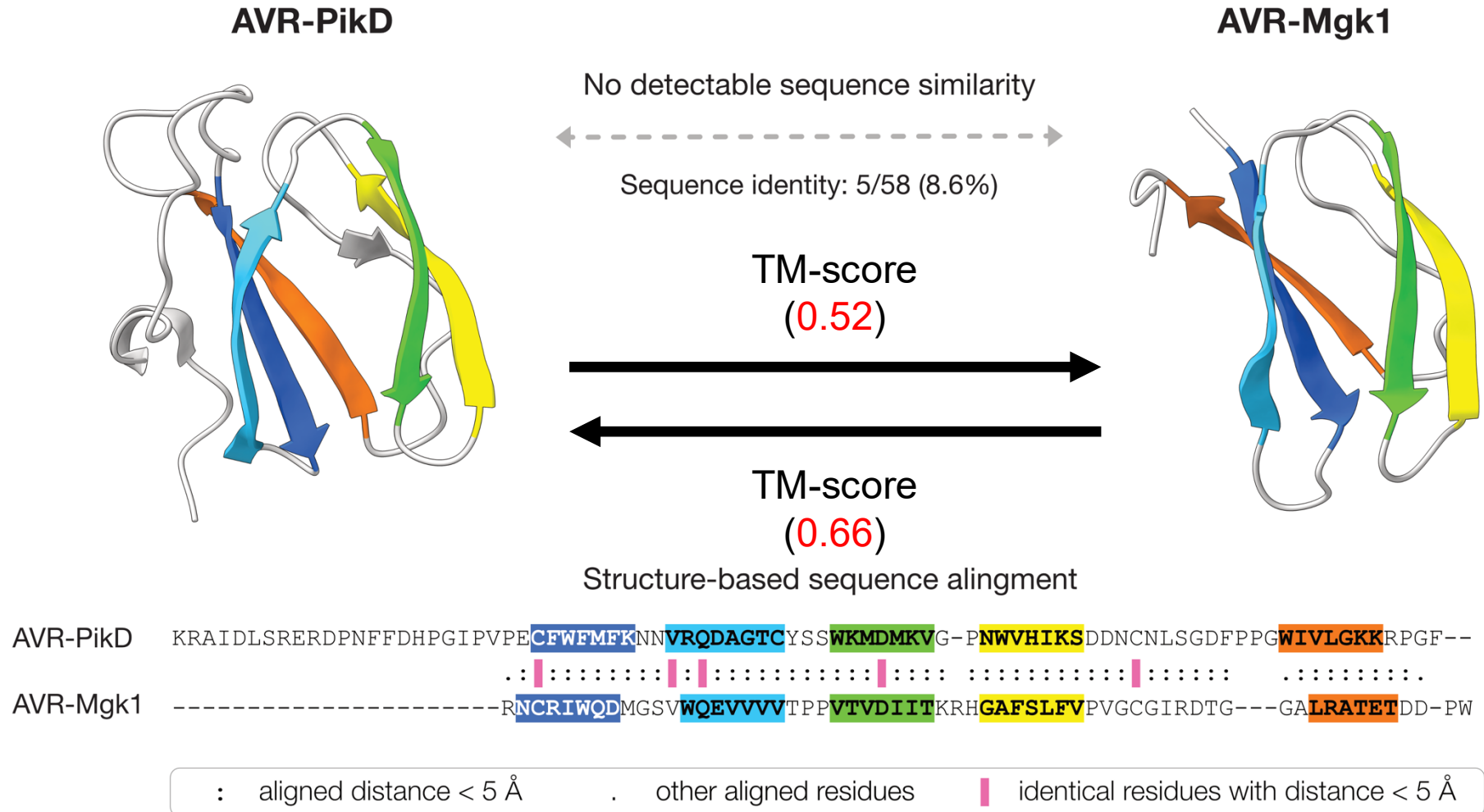
3_AVR-PikD



4_AVR-Mgk1



TM-align reports two scores, each normalized by a different structure

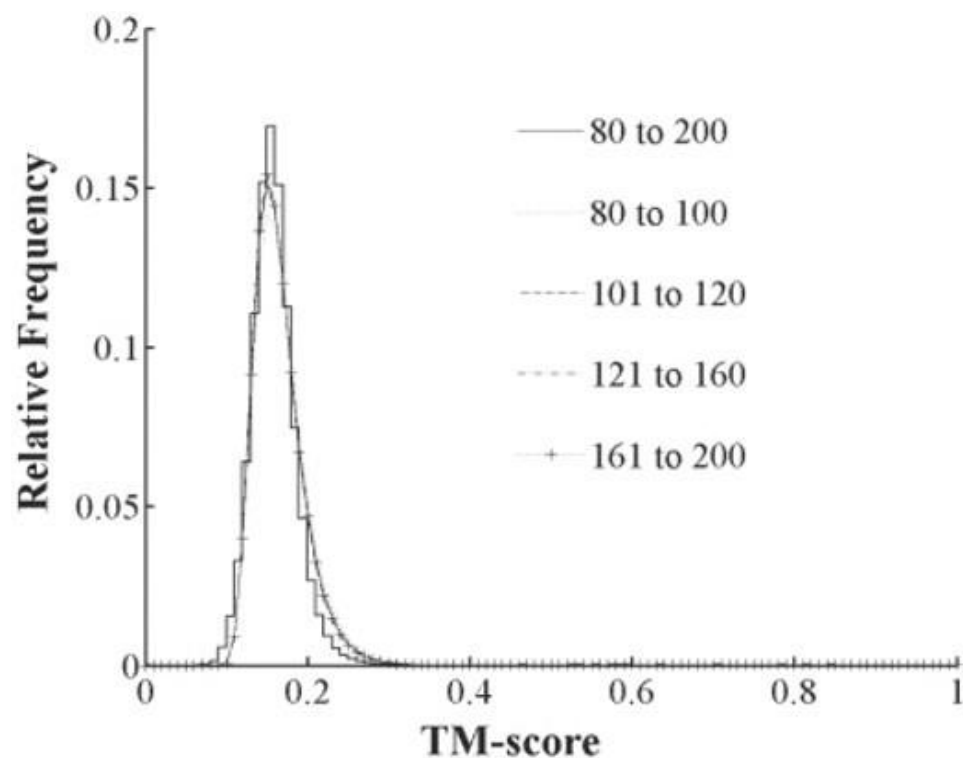


How significant is a protein structure similarity with TM-score = 0.5?

[Jinrui Xu](#), [Yang Zhang](#)  [Author Notes](#)

Bioinformatics, Volume 26, Issue 7, April 2010, Pages 889–895, <https://doi.org/10.1093/bioinformatics/btq066>

Published: 17 February 2010 **Article history** ▼

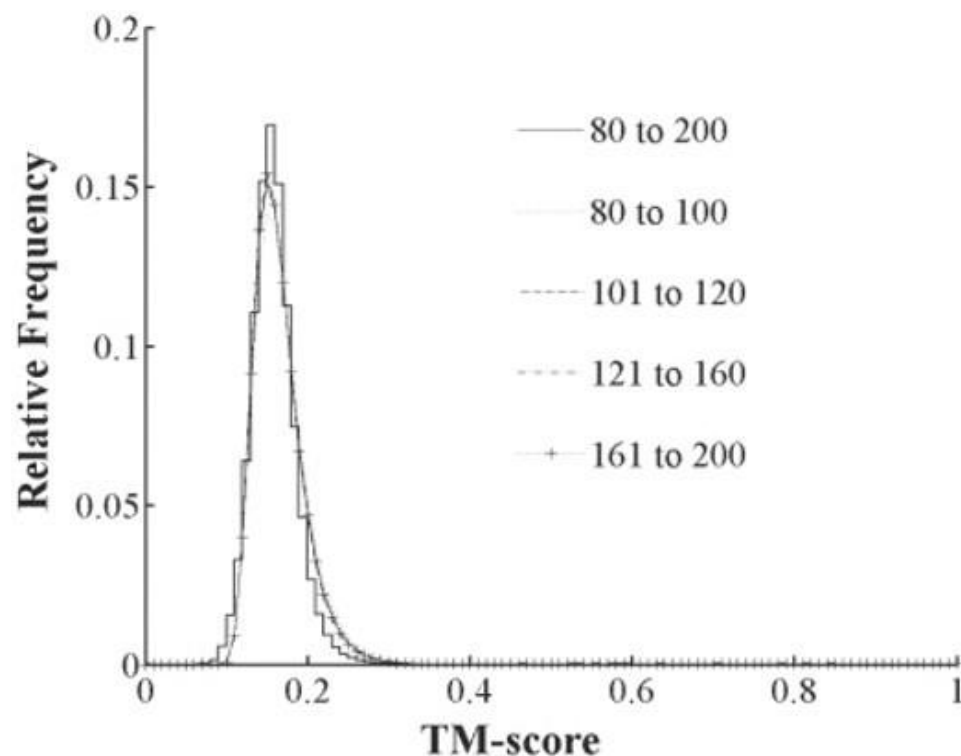


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"TM-scores below 0.2 correspond to randomly chosen unrelated proteins while those higher than 0.5 assume generally the same fold."

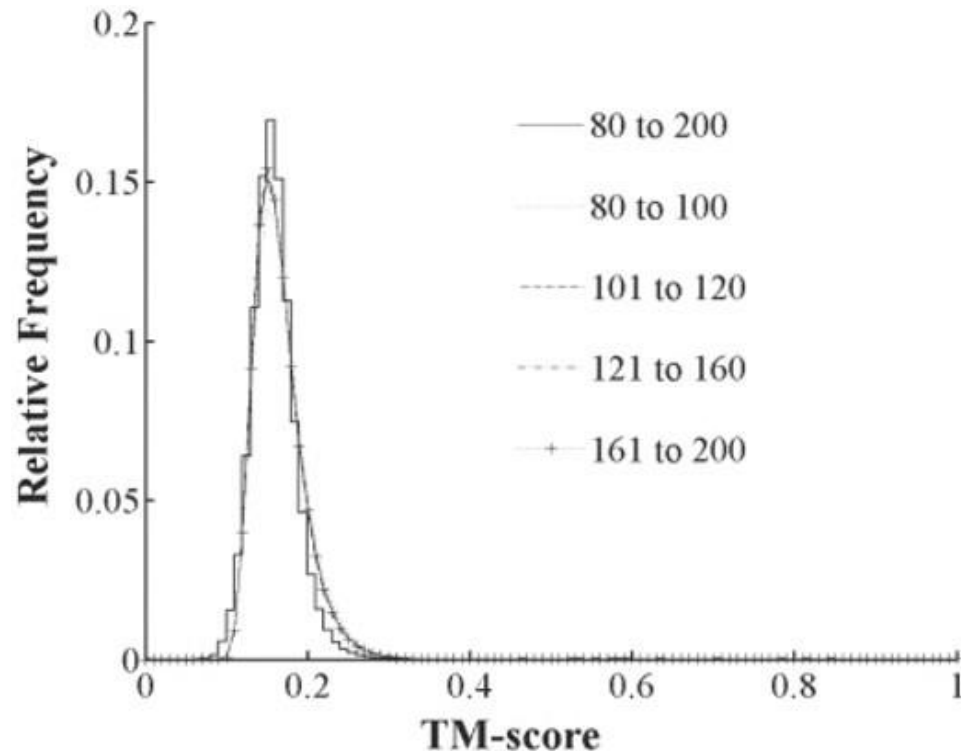
<https://zhanggroup.org/TM-align/>

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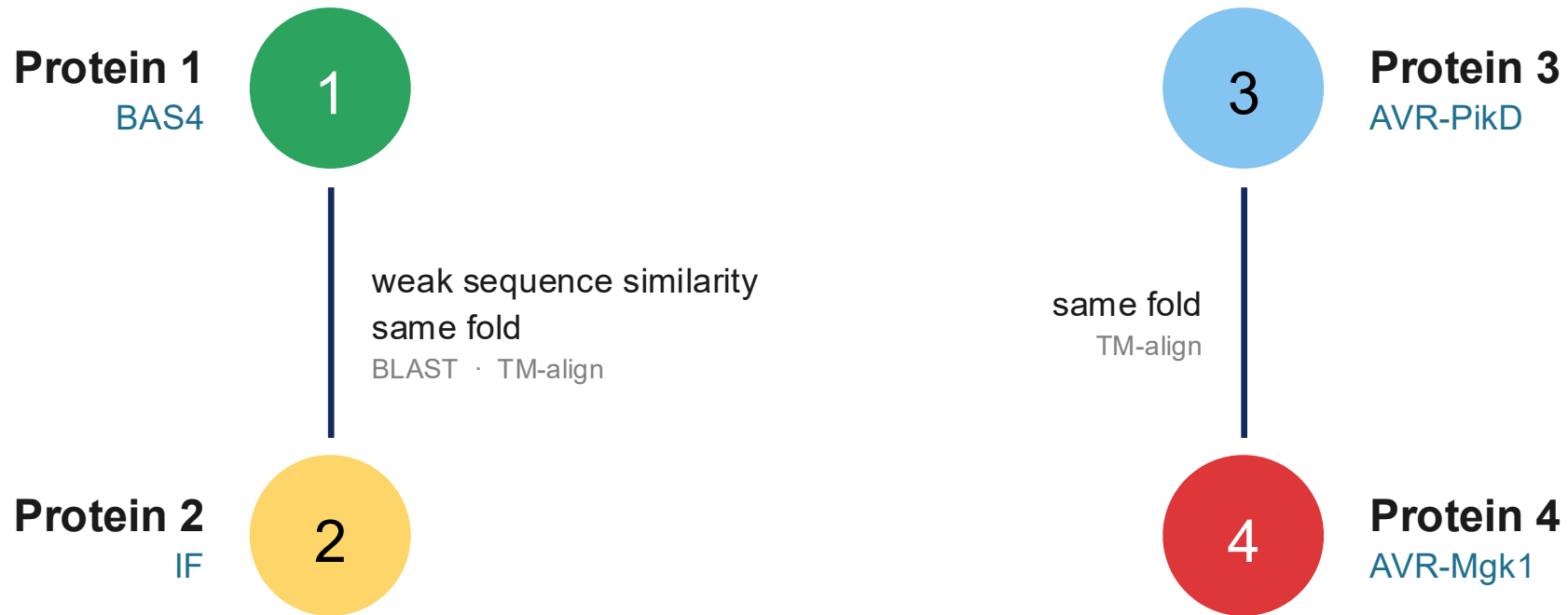
"TM-scores below 0.2 correspond to randomly chosen unrelated proteins while those higher than 0.5 assume generally the same fold."

<https://zhanggroup.org/TM-align/>

RCSB PDB performs the same pairwise alignment
<https://www.rcsb.org/alignment>

<https://www.rcsb.org/docs/tools/pairwise-structure-alignment#4-align-multiple-structures-to-an-alfafold-structure>

TM-align shows a shared fold between Protein 3 and Protein 4



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Exercise 3: Searching for similar structures with Foldseek

Exercise 4: Aligning many structures with FoldMason

Exercise 3: Searching for similar structures with Foldseek (Dataset 2)

Upload one of the structures from Dataset 2 to Foldseek:

<https://search.foldseek.com/search>

The screenshot shows the Foldseek Search web interface. At the top, there are four buttons: 'LOAD ACCESSION', 'UPLOAD PDB', 'PREDICT', and 'UPLOAD PREVIOUS RESULTS'. The 'UPLOAD PDB' button is circled in red with a '1' and has a red arrow pointing to it with the text 'Upload the PDB file'. Below these buttons is a section titled 'Databases & search settings'. Under 'Databases', there is a list of databases with checkboxes: AlphaFold/UniProt50 v6, AlphaFold/Swiss-Prot v6, AlphaFold/Proteome v6, CATH50 4.3.0, BFVD 2023_02_v2, MGnify-ESM30 v1, PDB100 20240101, GMGCL 2204, and BFMD 20240623. All are checked. Under 'Mode', there are three radio buttons: 3Di/AA, TM-align (selected), and LoL-align. To the left of the 'TM-align' radio button is a red circle with a '2' and the text 'Select TM-align'. Below the mode section is a 'Taxonomic filter' input field. At the bottom left, there is a checkbox for 'Iterative search'. At the bottom right, there is a blue 'SEARCH' button and a 'Summary' section that says 'Search all available databases with Foldseek in TM-align mode.' A red circle with a '3' and the text 'Click "SEARCH"' points to the 'SEARCH' button.

3Di/AA

local similarity, structure and sequence together

TM-align

overall (global) structural similarity

LoL-align

local structural similarity, a sign of homology

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Dataset 3: MAX-fold structures from Lahfa et al., 2024

RESEARCH ARTICLE

The structural landscape and diversity of *Pyricularia oryzae* MAX effectors revisited

Mounia Lahfa¹, Philippe Barthe¹, Karine de Guillen¹, Stella Cesari², Mouna Raji¹, Thomas Kroj², Marie Le Naour—Vernet², François Hoh¹, Pierre Gladieux², Christian Roumestand¹, Jérôme Gracy¹, Nathalie Declerck¹, André Padilla^{1*}

¹ Centre de Biologie Structurale, Univ Montpellier, CNRS UMR 5048, INSERM U 1054, Montpellier, France,
² PHIM Plant Health Institute, Univ Montpellier, INRAE, CIRAD, Institut Agro, IRD, Montpellier, France

<https://doi.org/10.1371/journal.ppat.1012176>

MAX group	representative structure	MAX ID	SS(*)	Structural specificities and extensions	MAX group	representative structure	MAX ID	SS(*)	Structural specificities and extensions
A		MAX39, MAX40, MAX41, MAX42, MAX109	1	additional β7β8 strands flanking one side of the β2β1β6 sheet and C-terminal extensions after β8 that are in helical conformations	I		MAX67, MAX80, MAX91, MAX103	1	long β1 strand paired with long β2 or β6 strands
B		MAX12, MAX13, MAX14	1	additional β7β8 strands flanking one side of the β2β1β6 sheet	J		MAX10, MAX11	1	presence of a short helix in the connecting loop between β5 and β6
C		MAX08, MAX09	1	presence of the β7 additional strand running antiparallel to the β6	K		MAX47, MAX92	1, 4	disulfide bond "4" joining the N-terminus of β1 and the C-terminal extension
D		MAX25, MAX53, MAX75	1	presence of the β7 additional strand running antiparallel to the β6	L		MAX34, MAX48	1, 3	poorly folded β2 strand and anti-parallel β4β5 strands stabilized by a disulfide bond
E		MAX57, MAX83	1	long β1 and β6 strands, C-terminal helical extensions within the MAX core	M		MAX15, MAX30, MAX76, MAX101, MAX111	1	long β1 and β2 strands and their connecting loops
F		MAX29, MAX64	1	C-terminal helical extensions that are stabilized by local residue-specific interactions (disulfide bond or hydrophobic residues)	N		MAX35, MAX63, MAX69, MAX88, MAX102, MAX108	1	unusually long loop making the connection between strands β2 and β3
G		MAX32, MAX90	1, 3	long loop joining the β1 and β2 strands and long N- and C-terminal extensions with putative helices of variable length and orientation	O		MAX21, MAX82, MAX70, MAX110	1	Similar β3β4β5 sheet
H		MAX52A, MAX52B	1, 3	two homologous MAX domains with conserved di-SS pattern	AVR-Pib		MAX61, MAX62, MAX05, MAX19, MAX72	[1]	MAX61 and MAX62 do not have the conserved SS1
AVR-Pia		MAX58, MAX93	1	β2 strand interaction with HMA (**)	AvrPiz-t		MAX06, MAX07	1, 3	very similar structures with the main difference for the β3β4 connecting loop
AVR-Pik		MAX01, MAX02, MAX03	1	β3 strand and N-terminal extension interacting with HMA (**)	MoToxB		MAX24, MAX33, MAX46, MAX89	1, 2	Structurally similar to ToxB with disulfide bond "2"

Exercise 4: Aligning many structures with FoldMason (Dataset 3)

Upload the structures from Dataset 3 to FoldMason:
<https://search.foldseek.com/foldmason>

≡

FoldMason MSA



 Search

 Multimer search

 FoldMason MSA

 Foldisco search

 Riboseek RNA search

GITHUB

STEINEGGER LAB

Input protein structures (PDB/CIF) for MSTA

 CLICK TO SELECT FILES

Drag & drop at least two PDB/CIF files here.

①

Upload the PDB files



LOAD ACCESSION

UPLOAD PREVIOUS RESULT (JSON)

LOAD EXAMPLE MSTA

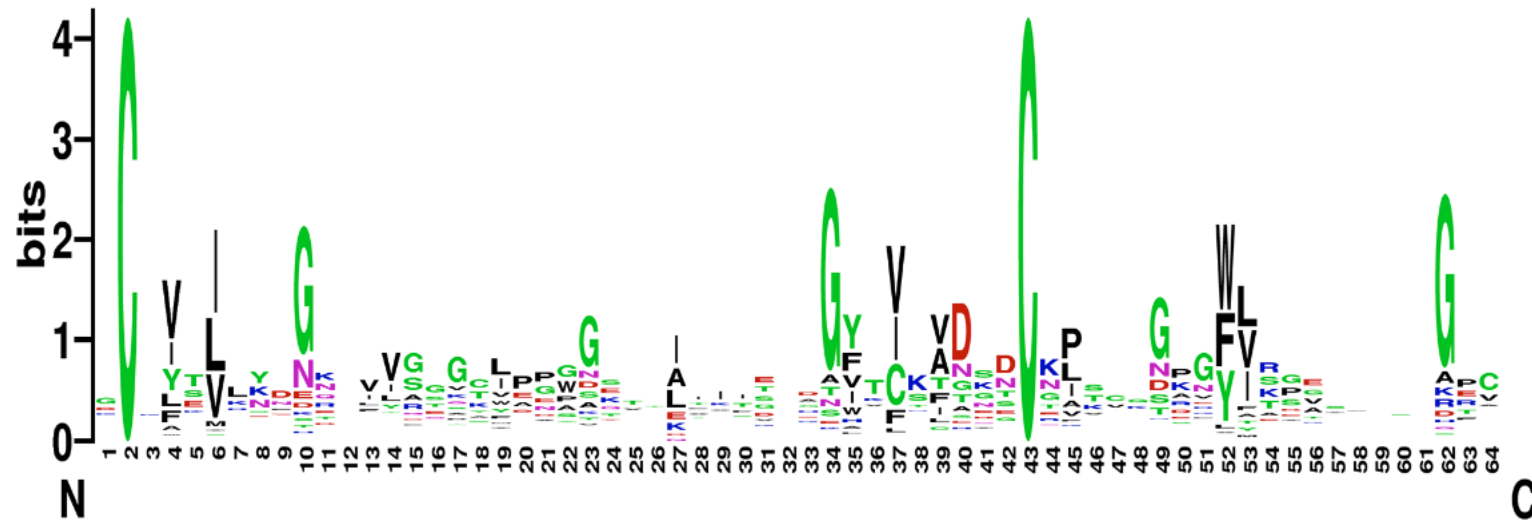
Click "ALIGN" ②

 ALIGN

Summary

Align 0 structures with FoldMason

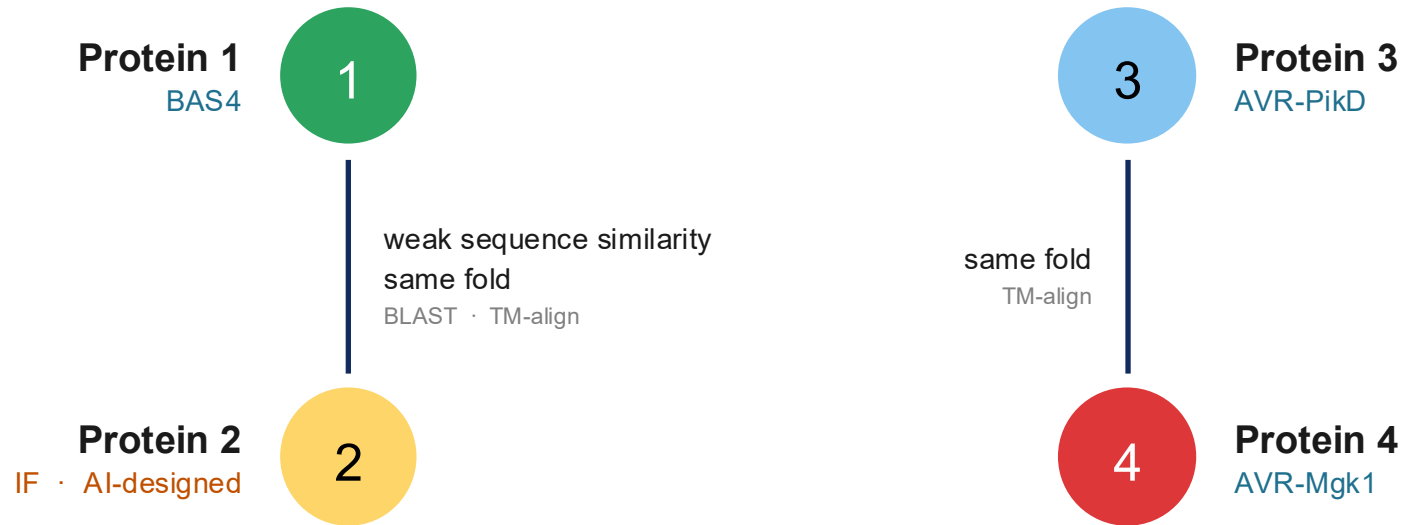
Two cysteine residues are conserved across MAX-fold proteins



Key points when interpreting structural similarity

1. Get everything you can from the sequence first.
2. The result depends on how good the input structure is.
3. Always look at the superposition, not just the score.

2_IF is not a natural protein: it was designed by inverse folding



Designed with ProteinMPNN

https://colab.research.google.com/github/dauparas/ProteinMPNN/blob/main/colab_notebooks/quickdemo.ipynb

Score	Expect	Method	Identities	Positives	Gaps
30.8 bits(68)	9e-07	Compositional matrix adjust.	15/87(17%)	37/87(42%)	0/87(0%)
Query	2	LATGDLHSAVCVTKRVNFFGHKTFQISP NATMCACTFYKGRNTGTNPWDKCEDCKYDGL	61		
Sbjct	2	IEEGNYCCAGACISSVTVNGVVS YTKNIAGTKCACELMRALDSGSAKGDTAPNAYFNGE	61		
Query	62	KCTSENGTIGGDEFKYYCEKKCGAKGS	88		
Sbjct	62	NCLDPDCNLSYSRLRHFCVNECGCAGA	88		

Four things to take away

- 1 **Get everything you can from the sequence first.**

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- 2 **Structural similarity can reveal hidden relationships.**

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Four things to take away

- 1 **Get everything you can from the sequence first.**
- 2 **Structural similarity can reveal hidden relationships.**
- 3 **Your answer is only as good as the structures you put in.**
- 4 **No single tool gives the whole answer.**

Programme

The same two-day programme runs at both venues (Espace Zmorda, then ISBS).

Day 1 Predict & Interpret

08:30–09:30	Registration
09:30–09:45	Welcome & Introductions
09:45–10:15	Introduction to AI for Genomics
10:15–11:00	Theoretical Introduction to AlphaFold
11:00–11:20	Break
11:20–12:30	 Practical 1: Predict
12:30–13:30	Lunch
13:30–15:00	<i>GOHREP it!</i> Interactive discussions
15:00–15:15	Break
15:15–17:00	 Practical 2: Interpret
17:00–17:15	Q&A

Day 2 Compare & Design

09:00–09:30	Day 2 Welcome & Registration
09:30–11:15	 Practical 3: Compare
11:15–11:30	Break
11:30–12:30	Model your own proteins
12:30–13:30	Lunch
13:30–14:45	 Practical 4: Design
14:45–15:05	Break
15:05–16:15	Genomics in an AI world
16:15–16:30	Summary
16:30–17:00	Q&A
17:00	Meeting close